

Education

- Sep 2023–ongoing **École Polytechnique Fédérale de Lausanne**,
École doctorale de Mathématiques.
◦ PhD candidate in the group of Mathematical Analysis and Calculus of Variations, supervised by prof. M. Colombo and prof. D. Albritton (University of Wisconsin, Madison)
◦ **Research interests:** analysis of PDEs, instability and non-uniqueness phenomena in fluid dynamics.
- Sep 2021–Jul 2023 **École Polytechnique Fédérale de Lausanne**,
Computational Science and Engineering, Master of Science.
◦ Double Degree Program with Politecnico di Milano
◦ **Master Thesis:** “Forward self-similar solutions for the forced 3D Navier Stokes Equations”, supervised by prof. M. Colombo (EPFL) and prof. M. Grasselli (Politecnico di Milano).
- Jan 2021–Sep 2022 **Alta Scuola Politecnica**,
Politecnico di Milano, Excellence multidisciplinary path.
- Sep 2020–Jul 2023 **Politecnico di Milano**,
Mathematical Engineering, Master of Science, 110/110 *cum laude*.
- Sep 2017–Sep 2020 **Politecnico di Milano**,
Ingegneria Matematica, Bachelor of Science, 110/110 *cum laude*.

Publications

- Apr 2026 *Topological entropy is generically infinite for non-Lipschitz velocity fields*, Johansson, C., Mescolini, G., arXiv:2604.01077.
- Jul 2025 *Vanishing viscosity non-unique solutions to the forced 2D Euler Equations*, Albritton, D., Colombo, M., Mescolini, G., arXiv:2507.19257.
- Dec 2024 *Self-similar instability and forced nonuniqueness: an application to the 2D Euler equations*, Dolce, M., Mescolini, G., Journal of the London Mathematical Society (2025).
- Dec 2024 *On vanishing diffusivity selection for the advection equation*, Mescolini, G., Pitcho, J., Sorella, M., Annali di Matematica Pura ed Applicata (2025).

Invited Talks

- Jul 2026 *Fluids, Waves and Sand Beach*, Santa Margherita di Pula (IT).
- Jun 2026 *Singularities, non-uniqueness and complex dynamics of incompressible fluids*, University of Bath (UK).
- Apr 2026 *Swiss Mathematical Society Doctoral Day*, ETH Zurich (CH).
- Apr 2026 *FoStEring MATH taLEntS*, University of Freiburg (DE).
- Dec 2025 *Nonlinear PDEs Seminar*, Karlsruhe Institute of Technology (DE).
- Nov 2025 *Junior Analysis Seminar*, Imperial College (UK).
- Sep 2024 *PDE Geometric Analysis seminar*, University of Wisconsin–Madison (US).
- May 2024 *International Conference on Elliptic and Parabolic Problems*, Gaeta (IT).
- Apr 2024 *Bernoullis Tafelrunde*, University of Basel (CH).

Seminar Organization

- Sep 2025 Workshop: *New challenges in Hyperbolic problems* (with De Nitti, N. and Johansson, C.), Bernoulli Center, EPFL, Lausanne (CH).

Achievements

- Oct 2025 SB Dean's Teaching Award, EPFL.
- Sep 2023 Best Thesis Poster, EPFL.
- July 2021 Basic Sciences Fellowship, EPFL.
- Feb 2019 Best Freshmen Prize, Politecnico di Milano.

Participation to conferences

- Jan 2026 Workshop: *Turbulence and Mixing in Fluid Dynamics*, SwissMAP Research Station, Les Diablerets (CH).
- July 2025 Workshop: *3rd Simons Math Summer Workshop: Partial Differential Equations of Classical Physics*, Simons Center for Geometry and Physics, Stony Brook (USA).
- June 2025 Workshop: *Blurring the lines between pure and applied through mixing*, Sabhal Mor Ostaig, Isle of Skye (UK).
- Apr 2025 Workshop: *Coherent structures and (in)stabilities in fluids*, GSSI, L'Aquila (IT).
- May 2024 Summer school: *(In)-stability phenomena in fluids*, CY Cergy Paris Université, Paris (FR).
- May 2024 Conference: *International Conference on Elliptic and Parabolic Problems*, Gaeta (IT).
- Feb 2024 Winter school: *Phase mixing, kinetic theory and fluid mechanics*, SwissMAP Research Station, Les Diablerets (CH).
- Sep 2023 Workshop: *Enjoying Probability and Fluids in Lausanne*, EPFL, Lausanne (CH).
- Jul 2023 Summer school: *Exotic solutions and well-posedness in ODEs and PDEs*, RISM, Varese (IT).
- Jul 2023 Summer school: *Deterministic and random features of fluids*, EPFL, Lausanne (CH).
- Jun 2023 Summer school: *New trends in mathematical fluid-dynamics*, Institut Fourier, Grenoble (FR).

Teaching

- 2025 Introduction to PDEs (Fall), Analysis IV (Spring), EPFL, Teaching assistant.
- 2024 Introduction to PDEs (Fall), Analysis IV (Spring), EPFL, Teaching assistant.
- 2023 Calculus of Variations (Fall), Analysis IV (Spring), EPFL, Teaching assistant.

Outreach

- Fall 2025 Cours Mathilda, *Gymnase Cantonal de Nyon*. "Nœuds et turbulence" (French).
- Fall 2025 TecDays, *Lycée-Collège de la Planta (Sion)*. "Des curiosités mathématiques" (French).
- Spring 2024 Outreach workshop, *EPFL*. "Teoria dei nodi e equazioni differenziali" (Italian).
- Fall 2023 Journée des gymnasiens, *EPFL*. "L'inégalité isopérimétrique" (French).

Work Experience

- Aug 2022– **R&D Intern**, Nestlé Research, Lausanne.
- Feb 2023 ◦ Internship in Machine Learning for Digital Nutrition.

Extra-Curricular Activities

- Sep 2022 – **SIAM Chapter at EPFL**.
- Jan 2025 ◦ Vice-President (Sep 2023–Jan 2025), Treasurer (Sep 2022–Sep 2023)

Skills

- Languages Italian (mother tongue), English (C1), French (B2)
- Coding Python, C, C++, SQL, MATLAB, R, FreeFem++
- Frameworks TensorFlow, Keras, PyTorch
- Utilities Anaconda, Git, LaTeX, Jupyter Notebook, Docker, SQL Server Management Studio
- CFD Software COMSOL Multiphysics, Ansys Fluent, PHOENICS